BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: PR - XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Min | **Maximum Marks:** 18

Note: [Urdu instructions printed at top of paper — best-effort] ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ ایک کا جواب درست ہے۔ جواب میں ایک سے زیادہ دائرے بھرے ہوں تو جواب غلط تصور ہوگا۔

Q1. Self-inductance (L) in terms of magnetic flux (φ), no of turns (N) and electric current (I) can be written as _____.

(A) $L = N\varphi/I$
(B) $L = N\varphi I$
(C) $L = I\varphi/N$
(D) $L = N/(I\varphi)$

Q2. The ratio of power in AC circuit to that of DC circuit is equal to _____.

(A) Sin φ
(B) Cos φ
(C) Tan φ
(D) Sec φ

Q3. Pair production can't take place in vacuum because _____ is not conserved.

(A) Momentum
(B) Charge
(C) Energy
(D) Mass

Q4. An electric charge q is placed at the centre of cube of side 'a' the electric flux on its one side will be _____.

(A) $q/(4\pi\varepsilon_0)$
(B) $q/(a^3\varepsilon_0)$
(C) $q/(6\varepsilon_0)$
(D) q/ε_0

Q5. $^{24}_{12}X = {}^{4}_{2}Y + ?? + Q$ The given reaction is completed by _____.

(A) Alpha (α)
(B) Beta (β)
(C) Gamma (γ)
(D) Sigma (δ)

Q6. By increasing applied anode voltage, wavelength (λ) of x-rays _____.

(A) Increases
(B) Decreases
(C) Remains unchanged
(D) Becomes infinite

Q7. The seismometer works on the principle of _____.

(A) Electromagnetism
(B) Electromagnetic induction
(C) Maxwell equations
(D) Ampere's Law

Q8. $^{238}_{92}U$ contains _____ number of nucleons.

(A) 146
(B) 238
(C) 92
(D) 330

Q9. For one complete cycle, the simple mean value of AC is _____.

(A) Zero
(B) Vm/2
(C) 2Vm
(D) 0.707 Vm

Q10. "mho" is a unit of _____.

(A) Resistivity
(B) Resistance
(C) Conductivity
(D) Conductance

Q11. The function of unipolar junction transistors depends on _____.

(A) Majority carriers
(B) Both A&B
(C) Minority carriers
(D) None of these

Q12. The super conductor has a _____ resistance.

(A) Moderate
(B) High
(C) Zero
(D) Infinite

Q13. How much voltage is required to make 2A flow through a resistance of 8Ω?

(A) 16 $A\Omega^{-1}$
(B) 16 ΩA^{-1}
(C) 16 $A\Omega$
(D) 16 $A^{-1}\Omega^{-1}$

Q14. Which of the given is a boson?

(A) Photon
(B) Neutrino
(C) Proton
(D) Pion

Q15. The reverse process of x-rays production can be related to _____.

(A) Compton effect
(B) Pair production
(C) Photoelectric effect
(D) None of these

Q16. Volt per meter is a non SI unit of _____.

(A) Electric force
(B) Voltage
(C) Electric power
(D) Electric intensity

Q17. Charge to mass ratio of electron is _____ that of proton.

(A) Equal to
(B) Less than
(C) Greater than
(D) Zero compare to

Q18. According to Wien's displacement law, with rise in temperature of black body, the product λmax T is _____.

(A) Increased
(B) Decreased
(C) Constant
(D) None of these

SECTION-B

Time Allowed: 2 Hours and 40 minutes

QQ2. Answer any TEN parts. Each part carries FOUR marks.

(1) Voltages are always measured between two points. Why?

(2) Why neutrons activated nuclides tend to decay by β− rather than β+?

(3) Determine electric field intensity due to a point charge.

(4) How much voltage is required to make 2A flows through an 8Ω resistance?

(5) A beam of red light and beam of blue light have exactly same energy. Which light contains greater number of photons?

(6) Explain AC power loss through capacitor and represent it graphically?

(7) Discuss the superconductivity of a conductor with the help of a curve?

(8) Can different equipotential lines cross each other?

(9) Define Alpha and Beta factor. Deduce the relation between α and β of a transistor.

(10) Differentiate between spontaneous emission and stimulated emission?

(11) Give the formula for the flux linkages in terms of angular orientation?

(12) What is an ammeter? How galvanometer is modified into ammeter?

(13) Describe the principle of metal detector.

SECTION-C

Note: Attempt any THREE questions. All questions carry equal marks.

QQ3.

(a) State and explain Ampere's circuital law. Determine Magnetic flux density of current carrying solenoid.

(b) Determine the wavelength of electron that has been accelerated through a potential difference of 50 volt.

QQ4.

(a) Explain PN junction as half wave and full wave rectifier.

(b) The full scale deflection for a galvanometer is 10mA. Its resistance is 100 Ω. How can it be converted into an ammeter of range 100A?

QQ5.

(a) State Photoelectric effect and explain it on the basis of photon theory of light?

(b) In common base circuit, the current gain is 0.96. If the collector current is 60 μA. Find the emitter current.

QQ6. Give a brief explanation of any TWO topics.

(a) What is Laser? Describe its principle and operation.

(b) Explain the principle, construction and working of mass spectrograph.

(c) State Gauss's law and show that [illegible in source scan]